Quantumtricity and the Smith Chart: First Report

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Summary

A Smith chart drawn from kronspice's AC analysis reads the energy levels of Kron's Schrödinger network directly: the H_2^+ ground state appears at −1.13841 Ha against the eigensolver's −1.13848 Ha on the same grid. The same chart, extended past the unit circle and to two-port and two-mode form, gives a place to measure negative resistance, torsion and a scalar field against the standard prediction.

- **Levels as resonances.** The impedance at one junction is the local Green's function. Each level visible there makes one full turn round the rim (12 levels, 12.9 turns in a nearly lossless sweep).
- **Hidden levels.** A level whose wave function is zero at the probe point does not appear. The 2pσu level (exact −0.6675 Ha) is missing from a midpoint probe, as it must be.
- **Particles as circuits.** On the light-ring picture (radius ħ/mc), every charged particle has $V/I = h/q^2$, the von Klitzing constant 25 812.807 Ω for charge e. This is an identity of the definitions, not a prediction. The fine-structure constant appears as an impedance ratio, $\alpha = (Z_0/4\pi)/(\hbar/e^2)$.
- **Extensions.** Negative resistance moves the trace outside the unit circle, and the ledger names who pays. Torsion breaks reciprocity (S12 ≠ S21). A scalar field adds a second wave, so reflection becomes a 2×2 matrix.
- **Status.** H22 stays *bounded*. Nothing here moves it yet; the tests in the last section could.

Everything in this report was run on a local copy of kronspice. No code has been committed.

Background: H22 in kronspice

Quantumtricity is register entry H22 in kronspice's CLAUDE.md: the Ohm's-law wheel (P, V, I, R) and the Standard Model wheel describe one structure, the quark model being electricity at the quantum level. Its status is *bounded*.

Step	What was tested	Result
1 (standard)	Circuit laws as the U(1) gauge theory on Kron's network	Passes. KVL is zero holonomy (2e-16); KCL is the rate of the Gauss generator (1e-17); the U(1) field on a Yee grid equals `em/fdtd.go` to 6.6e-13.
2 (hypothesis)	SU(3) link variables on the same graph	Null. SU(3) loops compose with defect 0.75; every circuit quantity composes with defect ⩽ 3e-16, the round-off of addition.

The standard prediction recorded beside it: circuit theory is the abelian, classical limit of a lattice gauge theory, and the quark model is the non-abelian one on the same kind of graph. The quantum units of V, I and R come from e and h through the Josephson, von Klitzing and single-electron standards, with no quark content.

The example's README lists three open items: a matter field on the nodes, where the wheel's terms land (R has no gauge-theory counterpart), and a TLM twin. This report adds a fourth: the Smith chart as the common instrument for reading circuits, molecules and particle models on one scale.

Smith chart on Kron's Schrödinger network

The chart found the H_2^+ ground state at −1.13841 Ha; the eigensolver gives −1.13848 Ha on the same 1,950-junction grid, a gap below one sweep step (about 3e-4 Ha).

Method. The circuit is `examples/kron_schrodinger/netlists/h2plus_coarse.cir`, Kron's 1945 network in its dual form. Its 1 A AC source at junction n960 (on the axis, between the nuclei) makes the junction voltage equal the driving-point impedance:

```
Z(\omega) = V_{n960}(\omega) / 1\,\mathrm{A}, \qquad \Gamma = \frac{Z - Z_0}
{Z + Z_0}, \qquad E = \frac{1}{\omega^2} - V_0
```

with V_0 = 5.08769 Ha, the network's potential shift. A shunt resistor C/τ at every junction gives each level a finite lifetime, an imaginary part of the energy. The sweep ran from E = −1.3 to +0.3 Ha in 4,000 points.

What the chart shows.

1. **Impedance is the local Green's function.** Re Z peaks at each level, with height proportional to $|\psi|^2$ at the probe junction.
2. **Levels hide at nodes.** No peak appears at the 2pσu level (exact −0.6675 Ha): its wave function is odd and vanishes at the midpoint. Moving the probe maps where each level lives.
3. **Counting levels.** With $\tau = 10^6$ the trace runs round the rim, 12.9 turns for 12 peaks found, one turn per visible level. This is a second check on the Sturm count used in `light_hydrogen`.
4. **Lifetime as circle size.** The loss knob sets each circle's Q = energy / width. This is the same picture particle physicists use for a resonance on an Argand diagram.

Level seen from n960	Chart (Ha)	Note
1sσg (ground)	−1.13841	Eigensolver on the same grid: −1.13848
next gerade level	−0.370	Coarse grid; finite differences sit below exact
2pσu	not seen	Node at the probe point (exact −0.6675)

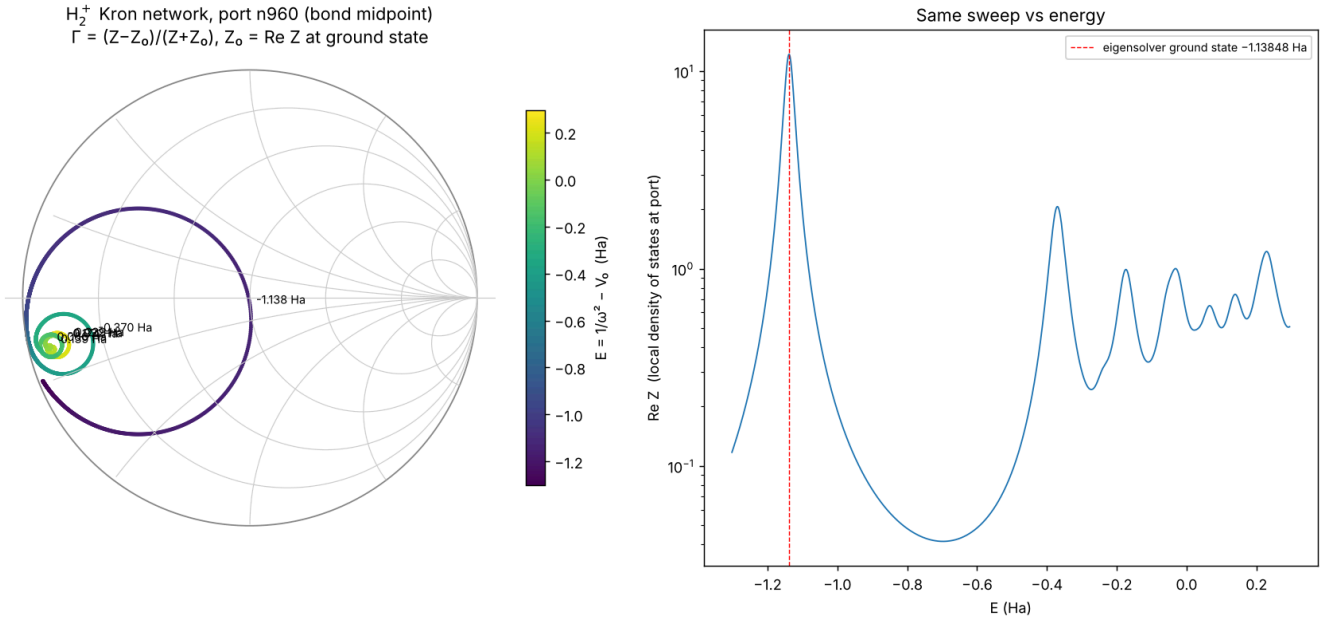


Figure 1. H_2^+ network seen from the bond midpoint: Γ over the sweep (left) and Re Z against energy (right).

Why the first chart sat on the left side

The trace sat on the left because the reference impedance Z_0 was set far above what the network shows almost everywhere. Nothing in the physics confines it there.

The left half of a Smith chart is exactly the region $|Z| < Z_0$:

```
\mathrm{Re}\,\Gamma = \frac{|z|^2 - 1}{|z + 1|^2}, \qquad z = Z / Z_0
```

Four things put the H_2^+ trace there:

1. **Z_0 was the ground-state peak, 12.19.** Away from that one peak, $|Z|$ is 0.3 to 1.3 (median Re Z 0.33). So $z \approx 0.03$ to 0.1, and Γ crowds toward the short-circuit point.
2. **The probe junction is stiff.** It is shunted by the rest of a 1,950-junction network, so off resonance it looks almost like a short.
3. **The dual form leans capacitive.** The circuit is Kron's network with capacitors and inductors exchanged. Off resonance its shunt capacitors dominate (median Im Z = −1.30), so the trace sits in the lower half.
4. **Loss shrinks the circles.** With $\tau = 400$, the smaller levels make small loops near the rim instead of full laps.

The side of the chart is a choice of normalisation and form. Each choice fills a different part of the chart:

Choice	Where the trace goes
Z_0 = median $\|Z\|$ (1.48)	Centre and lower half: the chart's centre sits where the data live
Nearly lossless ($\tau = 10^6$)	Full laps round the whole rim: top (inductive), right (open), bottom (capacitive), left (short)
Z_0 = each level's own peak	That level's circle passes through the centre (matched)
Kron's original all-inductor form	Impedance becomes admittance-like, $\Gamma \to -\Gamma$: the same trace mirrored to the right side

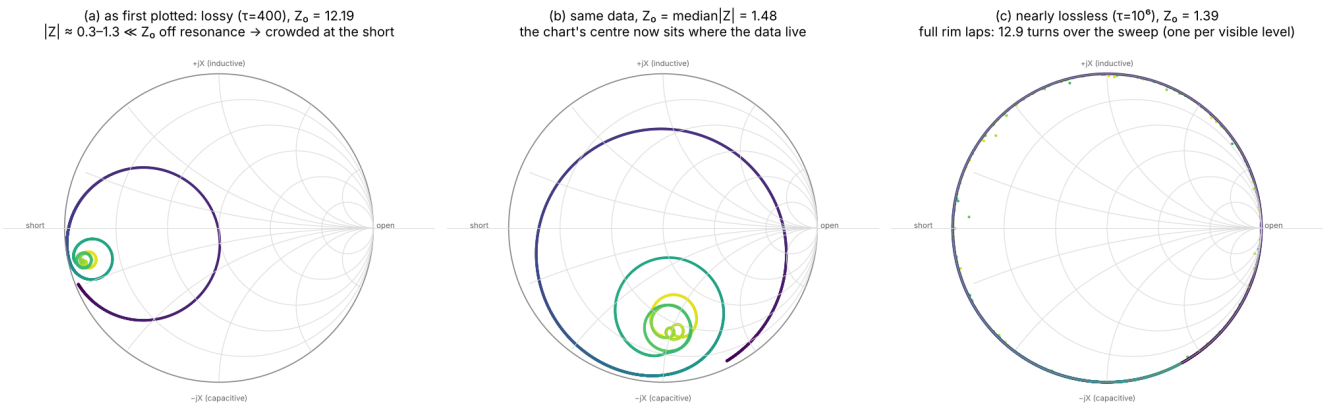


Figure 2. The same network three ways: (a) Z_0 = 12.19, crowded at the short; (b) Z_0 = median $|Z|$; (c) nearly lossless, full laps round the rim.

The particle table in the next section shows the opposite case. Seen from the vacuum's 30 Ω, a charge's $\hbar/q^2$ = 4,108 Ω is far larger, so every charged particle sits near the open-circuit point on the right.

Particles as circuits

On the light-ring picture of H23 and H31, every particle becomes an LC tank whose size and frequency are set by its mass, while its port impedance is set by its charge alone. The ring has radius $r = \hbar/mc$ and turns over once per Compton period:

```
f = \frac{mc^2}{h}, \quad C = 4\pi\varepsilon_0 r, \quad L = \frac{\mu_0 r}{4\pi}, \quad I = q f, \quad V = \frac{mc^2}{q}, \quad R_{\mathrm{loss}} = \frac{Z_0/4\pi}{Q}, \quad Q = \frac{mc^2}{\Gamma_{\mathrm{width}}}
```

Particle	q/e	f (Hz)	L (H)	C (F)	I (A)	V (V)	V/I (Ω)	R_loss (Ω)	Γ vs vacuum
e	−1	1.24e20	3.9e-20	4.3e-23	19.8	5.1e5	25 813	0 (stable)	0.9855
μ	−1	2.56e22	1.9e-22	2.1e-25	4.1e3	1.06e8	25 813	8.5e-17	0.9855
τ	−1	4.30e23	1.1e-23	1.2e-26	6.9e4	1.78e9	25 813	3.8e-11	0.9855
u	+⅔	5.22e20	9.1e-21	1.0e-23	55.8	3.2e6	58 079	confined	0.9935

Particle	q/e	f (Hz)	L (H)	C (F)	I (A)	V (V)	V/I (Ω)	R_loss (Ω)	Γ vs vacuum
d	−⅓	1.14e21	4.2e-21	4.7e-24	60.7	1.4e7	232 315	confined	0.9984
s	−⅓	2.26e22	2.1e-22	2.3e-25	1.2e3	2.8e8	232 315	confined	0.9984
c	+⅔	3.08e23	1.6e-23	1.7e-26	3.3e4	1.9e9	58 079	confined	0.9935
b	−⅓	1.01e24	4.7e-24	5.2e-27	5.4e4	1.3e10	232 315	confined	0.9984
t	+⅔	4.17e25	1.1e-25	1.3e-28	4.5e6	2.6e11	58 079	0.25	0.9935
W	±1	1.94e25	2.5e-25	2.7e-28	3.1e6	8.0e10	25 813	0.78	0.9855
Z	0	2.20e25	2.2e-25	2.4e-28	0	∞	open	0.82	1
H	0	3.03e25	1.6e-25	1.8e-28	0	∞	open	0.0009	1
ν (0.05 eV assumed)	0	1.21e13	3.9e-13	4.4e-16	0	∞	open	≈ 0	1
γ, g	0	no rest mass	—	—	—	—	the line itself, Z_0 = 376.73	0	—

Masses and widths are PDG 2024/2025 values; quark masses are scheme-dependent, and the Higgs width is the Standard Model prediction. Neutrino masses are unknown; 0.05 eV is a placeholder near the oscillation lower bound.

Readings, with the caveat on each.

- **V/I = h/q² for every particle.** For charge e this is R_K = 25 812.807 Ω. It follows from E = hf and I = qf, so the mass cancels: an identity, not evidence.
- **The ring impedance is $Z_0/4\pi \approx 30\ \Omega$ for every particle.** L and C both scale as 1/m.
- **The fine-structure constant is an impedance ratio.** 30 Ω / (ħ/e² = 4,108 Ω) = α. Seen from the vacuum, $\Gamma = (1 - \alpha q^2)/(1 + \alpha q^2)$, so the power coupling is about $4\alpha q^2$.
- **The gap.** e, μ and τ are identical circuits here, so the picture does not explain the 1 : 207 : 3477 mass ratios. R_loss is copied from measured lifetimes, not predicted.
- **Neutral particles have no electrical port.** The photon is the transmission line itself. The gluon would need SU(3) on the links, which H22 step 2 found no circuit quantity can carry.

Hypotheses already in the register, by particle.

Particle	Register entries	Gap
electron	H13, H21, H23, H24, H29, H44	—
neutrinos	H7, H28, H39–H42, H45	—
u, d (nucleons)	H22, H31	—
W, Z	H45 (weak force as the ring's own field)	—
photon	H23, H29, H30	—
gluon	H22 step 2	Null so far
μ, τ	none	"Heavy electrons" as smaller rings
s, c, b, t	none	Heavier quark rings
Higgs	none	A scalar field (closest: H8, H35)

Extending the chart for scalar, torsion and negative-resistance terms

Each new term needs its own extension of the chart, and each is drawn beside the standard (Maxwell) trace, with the difference $\Delta\Gamma$ as the measured quantity.

Term	Status in kronspice	Chart extension	Energy rule on the chart
Negative resistance	Built: `ModelSupplied` ledger column (`analysis/negres_test.go`)	Extended chart past $\|\Gamma\| = 1$, negative-r circles outside	$\|\Gamma\|^2 - 1$ = power supplied; must match a declared column
Torsion	Built: Hoffmann torsion (`kron/hoffmann.go`), Einstein 1928 (`examples/einstein_teleparallel`)	Two-port chart: S11, S22, and S21/S12	S21/S12 = 1 for a reciprocal circuit

Term	Status in kronspice	Chart extension	Energy rule on the chart
Scalar field S	Planned only (CLAUDE.md §3k); no `em/scalar/` yet	2×2 reflection matrix: ordinary and scalar wave	Largest singular value ⩽ 1 unless a column pays

Negative resistance

A kronspice series RLC (1 μH, 1 nF, Z_0 = 50 Ω) reached |Γ| = 2.33 with R = −20 Ω; R = 0 stayed on the rim to 1e-14 and R = +20 Ω stayed inside.

- **Use the extended chart, not the folded one.** RF practice often maps Γ → 1/Γ* and reads the circles as negative. In the run this puts the −20 Ω trace exactly on the +20 Ω trace, so the sign of the energy flow is lost.
- **Colour every point outside the circle by its paying column.** Candidates: a declared negative element, a pump (H16), a shaft (torsion), zero-point noise (H24, H28). A point outside with no paying column is a flaw to investigate, never credited as surplus.
- **Oscillation.** Where Γ_device · Γ_load = 1 the circuit oscillates (Kurokawa). This gives a ready test for H16, `negative_resistance_gain` and `barrett_osc`.

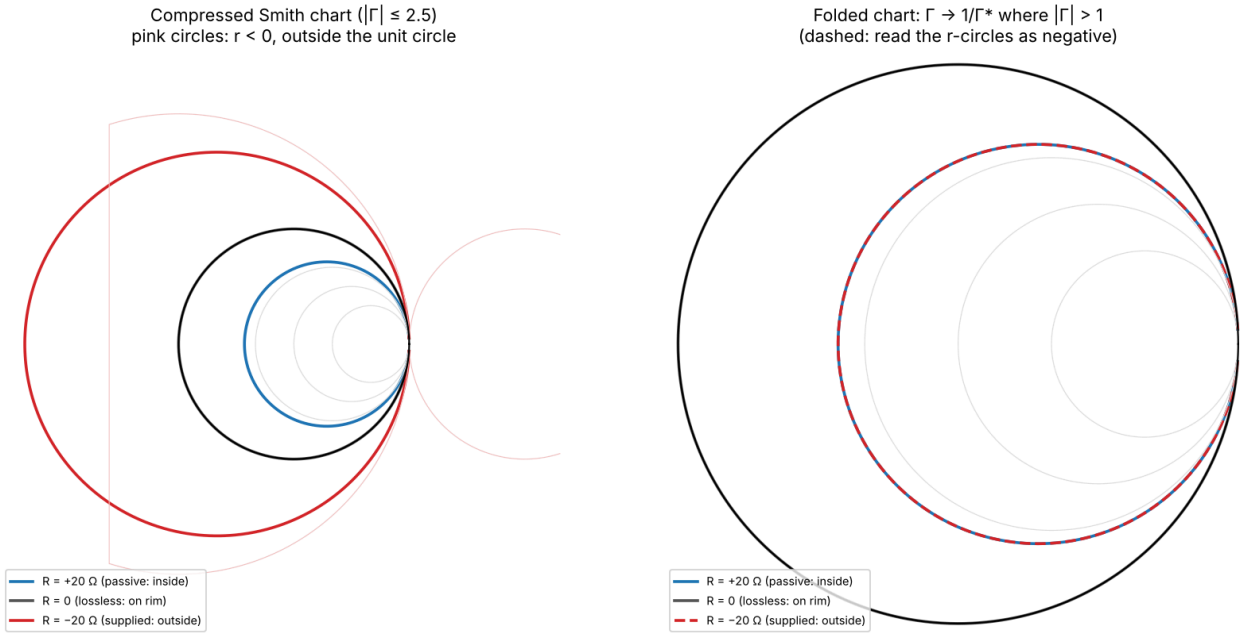


Figure 3. Series RLC with R = +20, 0, −20 Ω: extended chart (left) keeps the sign; folded chart (right) puts −20 Ω on top of +20 Ω.

Torsion

In Kron's machines, torsion appears as the speed-voltage term G·ω, which makes the impedance matrix non-symmetric. A one-port chart cannot show that; the two-port chart can.

- **Calibration case.** An induction machine driven above synchronous speed has negative slip, so its rotor term R_2/s is negative. Its electrical trace leaves the unit circle, and the ledger shows the shaft paying for every watt. This is a known physical negative resistance with its source booked.
- **Einstein's torsion.** His torsion trace is the electromagnetic potential, and to first order his theory gives Maxwell. So his version should match the standard trace; any ΔΓ would come from higher-order terms.

Scalar field

Van Vlaenderen and Hively keep the scalar $S = (1/c^2)\partial\varphi/\partial t + \nabla\cdot A$ that the Lorenz gauge sets to zero. That adds a longitudinal wave, so each port carries two waves:

```
\begin{pmatrix} b_T \\ b_S \end{pmatrix} = \begin{pmatrix} \Gamma_{TT} &
\Gamma_{TS} \\ \Gamma_{ST} & \Gamma_{SS} \end{pmatrix} \begin{pmatrix} a_T \\
a_S \end{pmatrix}
```

- **Diagonal entries.** Each wave's own reflection, on its own chart, normalised to its own impedance. The scalar wave's impedance must be derived from each model, not assumed equal to Z_0.
- **Off-diagonal entries.** Conversion between the two waves at a step or an antenna: the scalar effect itself.
- **Gates from §3k.** With the coupling at zero the off-diagonal entries vanish and the chart is ordinary (gate 1). The singular-value bound closes each model's energy theorem on the chart (gate 3).

A sink at the centre of the photon electron (H49)

The sink electron survives only if its inflow stays slower than the ether's waves: a localized loss in a superfluid ether does make a true horizon, but Earth's heat flow forbids the radiation that horizon would emit. This is `examples/ether_horizon`, numbered H49 because H47 and H48 are taken in the local register.

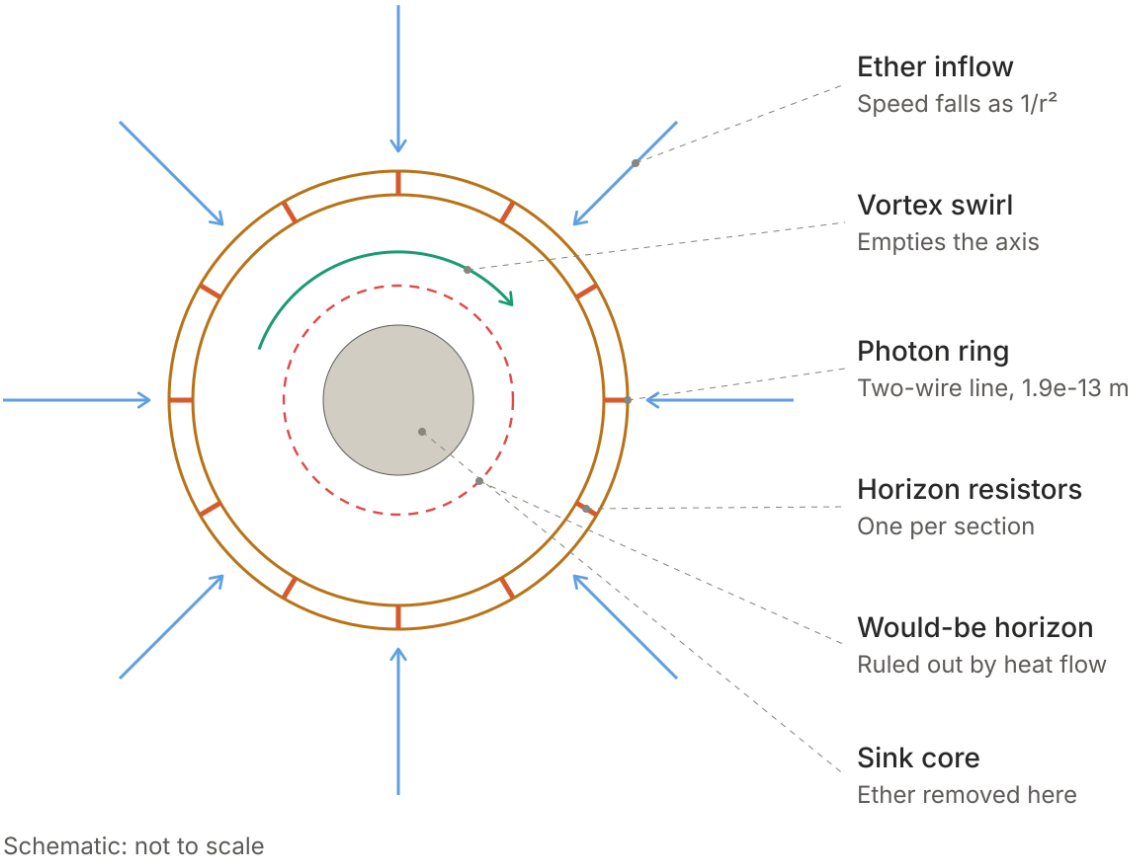


Figure 4. The sink electron in cross-section (schematic, not to scale).

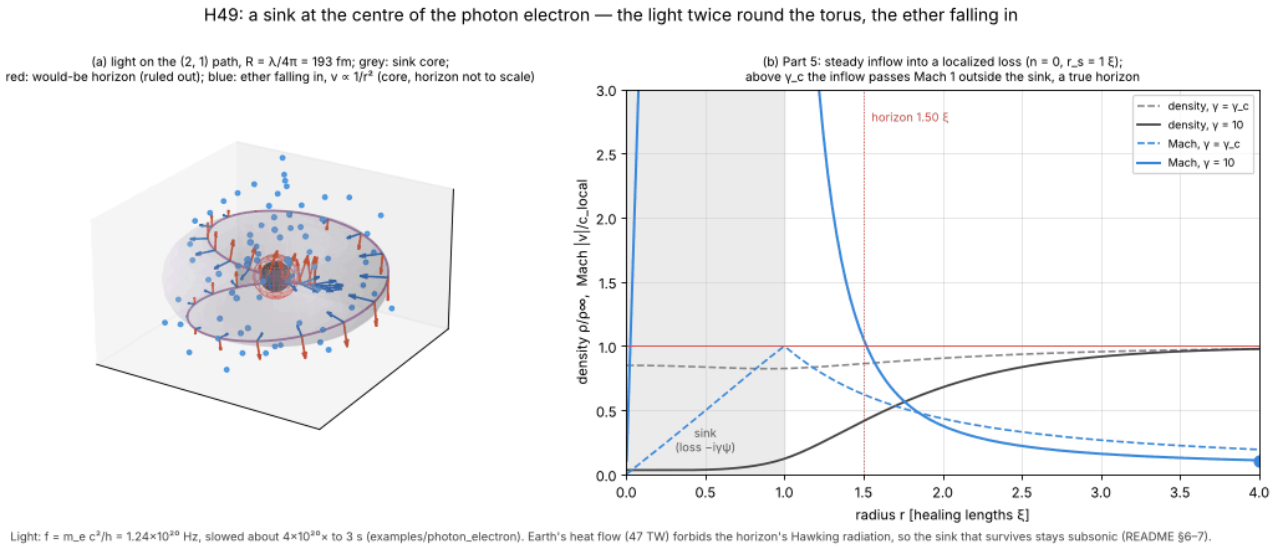


Figure 5. One period of the photon electron's light on its (2, 1) path, with the sink at the centre and ether falling in (left); the steady inflow into the sink at the critical loss and at γ = 10, horizon at 1.5 ξ (right).

- **Ether inflow.** Ether falls in from every side, its speed falling as $1/r^2$. Fitted to gravity, two sinks attract like masses, not like charges.
- **Vortex swirl.** A vortex empties its axis for every winding, but the void is a line one healing length thick, not a sphere.
- **Photon ring.** The light runs twice round the torus (H23). In kronspice it is the crossed two-wire ring line, with a horizon resistor across each section: kronspice reproduces the ladder's Q, and the energy ledger closes to 5×10^{-10}.
- **Would-be horizon.** A loss of about $\mu/\hbar$ makes the inflow pass Mach 1 just outside the sink. That horizon would have to be at least 2.2×10^{9} m across to keep its Hawking power under the heat bound, so it is ruled out.
- **Sink core.** The ether removed there is a ledger column the model still has to name: kept, it makes every body grow ($\rho_e \leqslant 1.2\times10^{-32}$ kg/m³ from the Sun's orbit data); destroyed, its energy leaves the books.

Tests and next steps

The single most discriminating test is a pulsating sphere: Maxwell says it radiates nothing, a scalar theory says it does, and a vector network analyser tells them apart.

The pulsating-sphere test. A sphere whose charge pulses evenly in all directions has no radiation in standard electromagnetism (Jefimenko; the electrical form of Birkhoff's theorem). After ohmic and feed-line losses are removed, its S11 must sit on the rim. A scalar-wave theory gives the pulsing mode a radiation resistance, which pulls the trace a set distance inside. It uses the ball antennas of the Monstein & Wesley setup already in §3k, and both predictions go on one chart.

Calibration cases, in order.

1. Negative resistor: $|\Gamma| > 1$ paid by `ModelSupplied`. Passes now.
2. Induction generator through the machine element: torsion, with the shaft paying.
3. H_2^+ level counting: one rim turn per visible level, checked against the Sturm count in `light_hydrogen`.
4. Pulsating sphere under each scalar model, after `em/scalar/` exists.

Code to add.

- [ ] `analysis/smith.go`: an N-port S-parameter sweep; each point carries the ledger column that pays for it; $|\Gamma| > 1$ allowed.

- [ ] A chart script: extended (never folded) chart, colour by ledger column, standard trace beside every hypothesis trace, ΔΓ reported.
- [ ] `em/scalar/`: the tier-2 interface of §3k, built against its four gates.
- [ ] `examples/particle_tanks`: the particle table above with tests, and a register entry beside H22.
- [ ] Radiation resistance of each particle ring (as H28 does for the neutrino ring), compared with the measured R_loss. This is the step that could turn the particle table from identities into predictions.

Sources

- kronspice: CLAUDE.md (H22, §3k), `examples/quantumtricity`, `examples/kron_schrodinger`, `examples/einstein_teleparallel`, `kron/hoffmann.go`, `analysis/negres_test.go`.
- PDG 2024 summary tables, gauge and Higgs bosons: W mass 80.3692 GeV, width 2.085 GeV.
- PDG 2025 review, mass and width of the W boson.
- G. Kron, "Electric circuit models of the Schrödinger equation," *Phys. Rev.* 67, 39 (1945).
- K. Kurokawa, "Some basic characteristics of broadband negative resistance oscillator circuits," *Bell Syst. Tech. J.* 48, 1937 (1969).
- Non-radiation of a spherically symmetric charge: by Gauss's law the field outside depends only on the total charge, so a pulsation with fixed total charge changes nothing outside (the electromagnetic analogue of Birkhoff's theorem).
- K. J. van Vlaenderen, arXiv:physics/0305098 (2003); L. M. Hively, *Int. J. Signals and Imaging Systems Eng.* 5(1), 3 (2012).
- G. Monstein and J. P. Wesley, *Europhys. Lett.* 59, 514 (2002).
- A. Einstein, *Sitzungsber. Preuss. Akad. Wiss.* 1928, 224.

Other particle masses, widths and lifetimes are standard PDG values from memory, rounded; they set only f, L and C, not the impedance results.